

Audit

1. Feature Overview

Business/security audit trail API and UI are **not** implemented (`com.bankone.audit` empty). Partial infra: JPA auditing via `AuditableEntity` + `JpaAuditingConfig` on some entities; Account/Customer use ad-hoc timestamp columns.

Status: Partial infrastructure / Stub product

2. Business Purpose

Answer who changed what and when for compliance and dispute resolution.

3. User Workflow

Planned: AUDITOR/ADMIN views audit log filtered by entity, user, date.

4. Execution Flow

Today: Spring Data auditing fills `created_by` / `updated_by` where `AuditableEntity` is used.

Planned:

```
AnyService.mutatingOperation()  
    ↓  
AuditService.record(event)  
    ↓  
AuditRepository.save()
```

5. Database Tables

No `audit_event` table. Auditing columns on `users`, `roles`, `user_roles`, `account_policy`.

6. REST APIs

None

7–13. Controllers / Services / Repositories / DTOs / Entities / Utils / Config

- Present: `AuditableEntity`, `JpaAuditingConfig` (common)
- Missing: audit module classes

14. Validation Rules

N/A

15. Security Rules

Planned: AUDITOR + ADMIN read; never public

16. Exception Handling

N/A

17. Logging

App logs ≠ audit trail (logs can rotate/lose data)

18. Audit Events

Planned event types: LOGIN_SUCCESS/FAIL, CUSTOMER_CREATE, ACCOUNT_OPEN, DEPOSIT, STATUS_CHANGE, POLICY_CHANGE, USER_UPDATE

19. Testing Strategy

When built: mutate entity → audit row; login fail → audit row

20. Future Extension Guide

Prefer explicit `AuditService` calls for financial events; keep JPA auditing for simple entity stamps.

Future Modification Guide

Requirement: Log every deposit

Item	Detail
Files	New <code>com/bankone/audit/**</code> ; <code>AccountServiceImpl.java</code>
Classes	<code>AuditService</code> , <code>AccountServiceImpl</code>
Methods	<code>deposit()</code> → <code>auditService.record("DEPOSIT", ...)</code>
Impact	Performance; storage; Reports overlap with Transaction

Requirement: Expose audit API

Item	Detail
Files	<code>AuditController</code> , <code>SecurityConfig</code> (AUDITOR), Angular reports/audit UI
Impact	Activate AUDITOR role currently seeded but unused

Call hierarchy (target)

```
AccountServiceImpl.deposit()  
    ↓  
AuditService.log()  
    ↓  
AuditRepository.save()
```